

Pre-deployment Preparation and Operations

This guide summarises Chapter 2 of the OceanGliders Best Practices document, led by Josh Kohut and Alvaro Lorenzo Lopez, with contributions from Nicole Waite, Estelle Dumont, Gillian Damerell, Dave Aragon, Steve Woodward, Victor Turpin, Adam Comeau, Theodore Thompson, and Tania Morales.

1. Pre-deployment Preparation

Glider deployments begin with a clear and communicated mission plan that enables proper planning and implementation throughout the mission – from pre-deployment preparation to recovery and post-deployment checks.

1.1 Mission Planning

All mission plans should start with a clear statement of the science and monitoring objectives, developed by the PI/Lead Scientist and communicated to the deployment team. At minimum, the plan should capture: required **track and/or area of operation, sensors, science sampling configuration, and mission duration**. The plan should be signed off by both the science lead and the glider logistics lead before any glider preparation begins.

Planning timelines are mission-dependent. The phases below are indicative minimums for straightforward missions and should be extended significantly for complex or international deployments.

More than 1 year before deployment

- Define the broad mission plan: area, timescales, glider behaviour (e.g. endurance line, station-keeping), sensor requirements, deployment/recovery options
- Assess major obstacles or show-stoppers: environmental, technical, operational, legal
- Discuss scientific sensor and depth requirements between researchers and the technical team
- Calculate endurance requirements accounting for duration/distance goals, payload, and environment (depth, stratification, currents)